Assignment 9 – Git Branching

Task 1: Observe the HEAD Pointer

The `git log --oneline` command was used to observe the commit history and identify the current position of the HEAD pointer.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git log --oneline
285cb17 (HEAD -> master) Update readme for branching assignment
b96ae7d Add readme file for branching assignment
bcd08b5 Remove readme file
3723d73 Add vector dot product program
81d612d Revert "Update project description"
e5baa53 Update project description
9533da2 Adding first file.
```

Task 2: Create a New Branch

The `git branch fast` command was used to create a new branch named `fast`, allowing development without affecting the stable version.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git branch fast
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git branch
  fast
* master
```

Task 3: Rename the Branch

The `git branch -m fast parallel` command was used to rename the branch from `fast` to `parallel`.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git branch -m fast parallel
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git branch
* master
  parallel
```

Task 4: Switch to the Parallel Branch

The `git checkout parallel` command was used to switch to the newly created branch. The `git status` command verified the current branch.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git checkout parallel
Switched to branch 'parallel'
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch parallel
nothing to commit, working tree clean
```

Task 5: Implement Parallel Dot Product

The Python program was modified to use the multiprocessing module for parallel computation of the dot product. The changes were committed to the parallel branch.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git log --oneline
7791726 (HEAD -> parallel) Implement parallel dot product algorithm
285cb17 (master) Update readme for branching assignment
b96ae7d Add readme file for branching assignment
bcd08b5 Remove readme file
3723d73 Add vector dot product program
81d612d Revert "Update project description"
e5baa53 Update project description
9533da2 Adding first file.
```

Task 6: Fix a Bug in the Main Branch

The main branch was checked out, and the vector size validation was modified to handle invalid input. The changes were committed, and the commit graph was displayed.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git --no-pager log --oneline --all --graph
* a57b857 (HEAD -> master) Fix invalid vector size validation
| * 7791726 (parallel) Implement parallel dot product algorithm
|/
* 285cb17 Update readme for branching assignment
* b96ae7d Add readme file for branching assignment
* bcd08b5 Remove readme file
* 3723d73 Add vector dot product program
* 81d612d Revert "Update project description"
* e5baa53 Update project description
* 9533da2 Adding first file.
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch master
nothing to commit, working tree clean
```

Task 7: Merge the Parallel Branch

The `git merge parallel` command was used to merge the parallel implementation into the main branch. Git detected conflicting changes because the same line was modified differently in both branches.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git merge parallel
Auto-merging vecadd.py
CONFLICT (content): Merge conflict in vecadd.py
Automatic merge failed; fix conflicts and then commit the result.
```

Task 8: Resolve Merge Conflicts

The conflicting sections were identified and manually resolved in VS Code. The required code was retained, conflict markers were removed, and the resolved file was committed to complete the merge.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git add vecadd.py
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git commit -m "Merge parallel branch and resolve conflicts"
[master b80d56a] Merge parallel branch and resolve conflicts
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git --no-pager log --oneline --all --graph
*   b80d56a (HEAD -> master) Merge parallel branch and resolve conflicts
|\
| * 7791726 (parallel) Implement parallel dot product algorithm
* | a57b857 Fix invalid vector size validation
|/
* 285cb17 Update readme for branching assignment
* b96ae7d Add readme file for branching assignment
* bcd08b5 Remove readme file
* 3723d73 Add vector dot product program
* 81d612d Revert "Update project description"
* e5baa53 Update project description
* 9533da2 Adding first file.
```

Task 9: Delete the Parallel Branch

The `git branch -d parallel` command was used to delete the parallel branch after successfully merging its changes into the main branch.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git branch -d parallel
Deleted branch parallel (was 7791726).
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git branch
* master
```

Task 10: Explore Git Stash

The `git stash` command was explored to temporarily save uncommitted changes. The `git stash push` command saved the modifications, while `git stash pop` restored them to the working directory.

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git stash push -m "My temporary readme changes"
Saved working directory and index state On master: My temporary readme changes
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git stash list
stash@{0}: On master: My temporary readme changes
```

```
PS C:\Users\Aryan\OneDrive\Documents\assign8-git> git stash pop
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   readme.txt

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (66684ce337e3a495c0cfe18c1093583c0ea4a1a5)
```